

Sundhedsdata og Interoperabilitet

Data quality + Interoperability

Feb 23, 2026

Chapter 2, Tim Benson & Grahame Grieve; Principles of Health Interoperability

Mohammed et al. 2025, The Five Facets of Data Quality Assessment

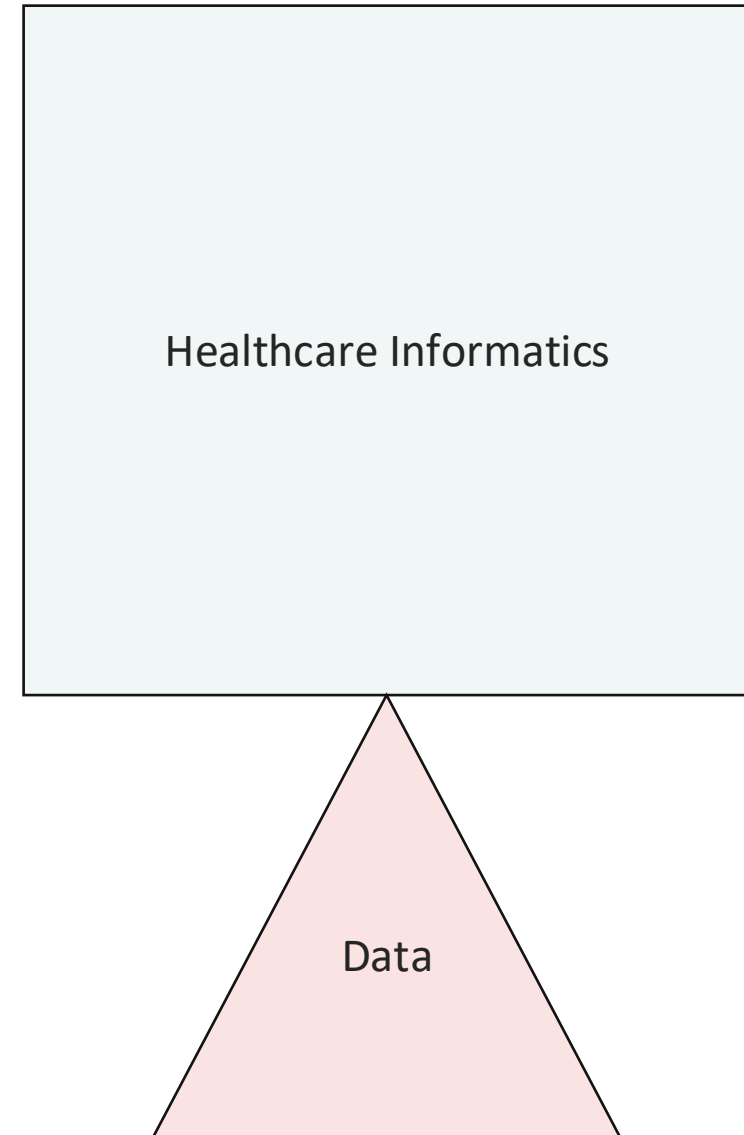
<https://doi.org/10.1145/3749116.3749120>

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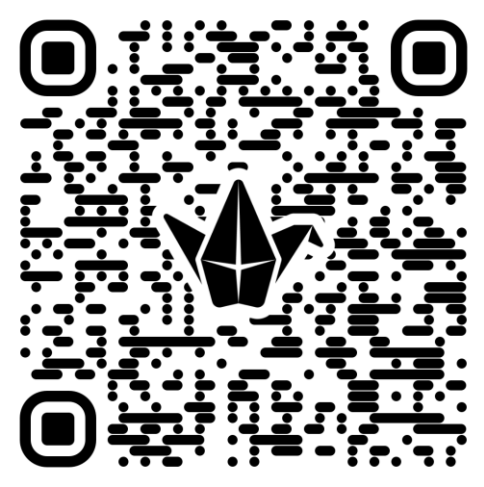
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Data quality

What does high-quality data mean to you?



What does data quality mean?

Good question, good answers!

Many people would agree, or not



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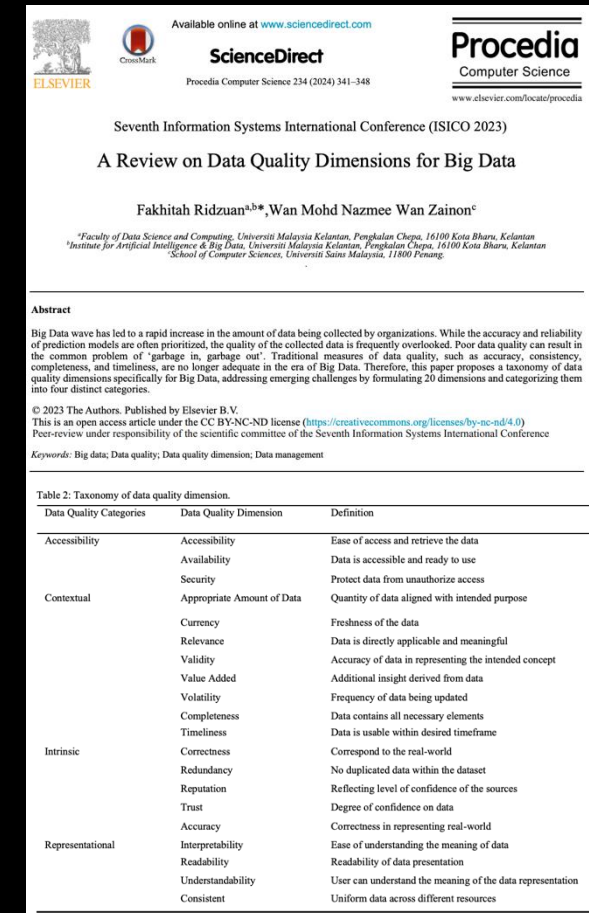


Table 2: Taxonomy of data quality dimension.

Data Quality Categories	Data Quality Dimension	Definition
Accessibility	Accessibility	Ease of access and retrieve the data
	Availability	Data is accessible and ready to use
	Security	Protect data from unauthorized access
Contextual	Appropriate Amount of Data	Quantity of data aligned with intended purpose
	Currency	Freshness of the data
	Relevance	Data is directly applicable and meaningful
	Validity	Accuracy of data in representing the intended concept
	Value Added	Additional insight derived from data
	Volatility	Frequency of data being updated
	Completeness	Data contains all necessary elements
Intrinsic	Timeliness	Data is usable within desired timeframe
	Correctness	Correspond to the real-world
	Redundancy	No duplicated data within the dataset
	Reputation	Reflecting level of confidence of the sources
	Trust	Degree of confidence on data
Representational	Accuracy	Correctness in representing real-world
	Interpretability	Ease of understanding the meaning of data
	Readability	Readability of data presentation
	Understandability	User can understand the meaning of the data representation
	Consistent	Uniform data across different resources

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G19

Data Quality Assessment

LEO L. PIPINO, YANG W. LEE, AND RICHARD Y. WANG

Ensuring data quality is a critical task for organizations. This paper discusses the importance of data quality and provides a framework for assessing data quality. The framework is based on the following dimensions: accessibility, appropriateness, believability, completeness, conciseness, consistency, ease of manipulation, free of error, interpretability, objectivity, relevancy, reputation, security, timeliness, understandability, and value-added.

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How good is a company's data quality? Answering this question requires usable data quality metrics. Currently, most data quality measures are developed on an ad hoc basis to solve specific problems [6, 8], and fundamental principles necessary for developing usable metrics in practice are lacking. In this article, we describe principles that can help organizations develop usable data quality metrics.

Studies have confirmed data quality is a multi-dimensional concept [1, 2, 6, 9, 10, 12]. Companies must deal with both the subjective perceptions of the individuals involved with the data, and the objective measurements based on the data set in question. Subjective data quality assessments reflect the needs and experiences of stakeholders: the collectors, custodians, and consumers of data products [2, 11]. If stakeholders assess the quality of data as poor, their behavior will be influenced by this assessment. One can use a questionnaire to measure stakeholder perceptions of data quality dimensions. Many healthcare, finance, and consumer product companies have used one such questionnaire, developed to assess data quality dimensions listed in Table 1 [7]. A major U.S. bank that administered the questionnaire found custodians (mostly MIS professionals) view their data as highly timely, but consumers disagree; and data consumers view data as difficult to manipulate for their business purposes, but custodians disagree [4, 6]. A follow-up investigation into the root causes of differing assessments provided valuable insight on areas needing improvement.

Objective assessments can be task-independent or task-dependent. Task-independent metrics reflect states of the data without the contextual knowledge of the application, and can be applied to any data set, regardless of the tasks at hand. Task-dependent metrics, which include the organization's business rules, company and government regulations, and constraints provided by the database administrator, are developed in specific application contexts.

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Journey to Data Quality

Table 1. Data quality dimensions.

Dimensions	Definitions
Accessibility	the extent to which data is available, or easily and quickly retrievable
Appropriate Amount of Data	the extent to which the volume of data is appropriate for the task at hand
Believability	the extent to which data is regarded as true and credible
Completeness	the extent to which data is not missing and is of sufficient breadth and depth for the task at hand
Concise Representation	the extent to which data is compactly represented
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Value-Added	the extent to which data is beneficial and provides advantages from its use

roadmap for implementing a viable data research and drawing on real-world

and structural. Neither ad hoc are or developing an expensive data journey to Data Quality offers a roadmap for implementing a viable data and on rigorous research and informed by and provides the principles, strategies, in the data quality field for many the importance of getting an or assessing data, both subjectively (by describe real problems and solutions, healthcare organization and data tting company policy on data quality ility.

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Seventh Information Systems International Conference (ISICO 2023)

A Review on Data Quality Dimensions for Big Data

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Abstract

Big Data wave has led to a rapid increase in the amount of data being collected by organizations. While the accuracy and reliability of prediction models are often prioritized, the quality of the collected data is frequently overlooked. Poor data quality can result in the common problem of 'garbage in, garbage out'. Traditional measures of data quality, such as accuracy, consistency, completeness, and timeliness, are no longer adequate in the era of Big Data. Therefore, this paper proposes a taxonomy of data quality dimensions specifically for Big Data, addressing emerging challenges by formulating 20 dimensions and categorizing them into four distinct categories.

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Peer-review under responsibility of the scientific committee of the Seventh Information Systems International Conference

Keywords: Big data; Data quality; Data quality dimension; Data management

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Representational	Interpretability	Ease of understanding the meaning of data
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G19 Data Quality Assessment

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Data quality assessment in healthcare, dimensions, methods and tools: a systematic review

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Elham Hosseinzadeh, Marziyeh Afkanpour, Mehri Momeni & Hamed Tabesh

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Frequency of dimensions

Readability
Understandability
Consistent

Readability of data presentation
User can understand the meaning of the data representation
Uniform data across different resources

Abstract

Accuracy and reliability of data quality can result in a taxonomy of data quality dimensions and categorizing them

Pharu, Kelantan
Bhara, Kelantan

What does data quality mean?

Data Quality Assessment

LEO L. PIPINO, YANG W. LEE, AND RICHARD Y. WANG

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There is a metric for everything
But do you need to use them all?

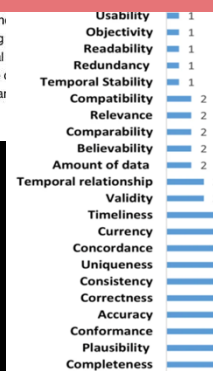
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Frequency of dimensions



Readability
Understandability
Consistent
Readability of data presentation
User can understand the meaning of the data representation
Uniform data across different resources

What about a more generic take?

Our best take yet on defining what is important when handling data

Technology agnostic – unlike, e.g., cookie banners

Focusing on outcomes and risks and not prescribing methods



What does data quality mean?

3. Training, validation and testing data sets shall be **relevant**, *sufficiently representative*, and *to the best extent possible*, **free of errors** and **complete** in view of the intended purpose. They shall have the **appropriate statistical properties**, including, where applicable, as regards the persons or groups of persons in relation to whom the high-risk AI system is intended to be used. Those characteristics of the data sets may be met at the level of individual data sets or at the level of a combination thereof.

4. Data sets shall *take into account, to the extent required by the intended purpose*, the characteristics or elements that are particular to the specific **geographical**, **contextual**, **behavioural** or **functional setting** within which the high-risk AI system is intended to be used.

Article 10 of the EU AI Act

<https://ai-act-service-desk.ec.europa.eu/en/ai-act/article-10>

What about a more generic take?



Relevant

sufficiently representative

to the best extent possible, free of errors and complete in view of the intended purpose.

Appropriate statistical properties met at the level of individual data sets or at the level of a combination thereof.

take into account, to the extent required by the intended purpose, the characteristics or elements that are particular to the specific geographical, contextual, behavioural or functional setting

Hospital_ID	Hospital_Name	Date	Time_Slot	Patients_Arrived	Avg_Wait_Minutes	Triage_Level_Mix
H001	Rigshospitalet	2024-03-15	08:00	12	45	L1:2, L2:4, L3:6
H001	Rigshospitalet	2024-03-15	12:00	18	67	L1:1, L2:8, L3:9
H001	Rigshospitalet	2024-03-15	16:00	23	89	L1:3, L2:10, L3:10
H002	Hvidovre Hospital	2024-03-15	08:00	8	34	L1:0, L2:3, L3:5
H002	Hvidovre Hospital	2024-03-15	12:00	15	52	L1:1, L2:5, L3:9
H002	Hvidovre Hospital	2024-03-15	16:00	19	71	L1:2, L2:7, L3:10
H003	Bispebjerg Hospital	2024-03-15	08:00	10	38	L1:1, L2:4, L3:5
H003	Bispebjerg Hospital	2024-03-15	12:00	14	58	L1:0, L2:6, L3:8
H003	Bispebjerg Hospital	2024-03-15	16:00	21	82	L1:2, L2:9, L3:10
H004	Amager Hospital	2024-03-15	08:00	7	29	L1:0, L2:2, L3:5
H004	Amager Hospital	2024-03-15	12:00	11	44	L1:1, L2:4, L3:6
H005	Frederiksberg Hospital	2024-03-15	08:00	9	41	L1:1, L2:3, L3:5
H005	Frederiksberg Hospital	2024-03-15	12:00	13	55	L1:0, L2:5, L3:8

Average emergency department (ED) wait times collected from 5 major hospitals in the Capital Region over the past 5 years, complete, no errors.

Hospital_ID	Hospital_Name	Date	Time_Slot	Patients_Arrived	Avg_Wait_Minutes	Triage_Level_Mix
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Average emergency department (ED) wait times collected from 5 major hospitals in the Capital Region over the past 5 years, complete, no errors.

Is it good quality data? Why? Why not?

Contextual data quality

It depends

Task: Medical resource reallocation for emergency services in the Capital Region.

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Average emergency department (ED) wait times collected from 5 major hospitals in the Capital Region over the past 5 years, complete, no errors.

Contextual data quality

It depends

Task: Medical resource reallocation for emergency services in the Capital Region.

- ✓ **Accuracy:** Data accurately reflects actual wait times
- ✓ **Representativeness:** The dataset covers the specific hospitals and patient population they're managing
- ✓ **Relevance:** Wait times directly inform staffing and resource decision
- ✓ **Completeness:** Complete records from the past 5 years account for differences in seasons, holidays, etc.

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Contextual data quality

It depends

Task: Design an app predicting which hospital to go to with an emergency for shortest wait time

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H003	Bispebjerg Hospital	2024-03-15	08:00	10	38	L1:1, L2:4, L3:5
H003	Bispebjerg Hospital	2024-03-15	12:00	14	58	L1:0, L2:6, L3:8
H003	Bispebjerg Hospital	2024-03-15	16:00	21	82	L1:2, L2:9, L3:10
H004	Amager Hospital	2024-03-15	08:00	7	29	L1:0, L2:2, L3:5
H004	Amager Hospital	2024-03-15	12:00	11	44	L1:1, L2:4, L3:6
H005	Frederiksberg Hospital	2024-03-15	08:00	9	41	L1:1, L2:3, L3:5
H005	Frederiksberg Hospital	2024-03-15	12:00	13	55	L1:0, L2:5, L3:8

Average emergency department (ED) wait times collected from 5 major hospitals in the Capital Region over the past 5 years, complete, no errors.

Contextual data quality

It depends

Task: Design an app predicting which hospital to go to with an emergency for shortest wait time

Hospital_ID	Hospital_Name	Date	Time_Slot	Patients_Arrived	Avg_Wait_Minutes	Triage_Level_Mix
H001	Rigshospitalet	2024-03-15	08:00	12	45	L1:2, L2:4, L3:6
H001	Rigshospitalet	2024-03-15	12:00	18	67	L1:1, L2:8, L3:9
H001	Rigshospitalet	2024-03-15	16:00	23	89	L1:3, L2:10, L3:10
H002	Hvidovre Hospital	2024-03-15	08:00	8	34	L1:0, L2:3, L3:5
H002	Hvidovre Hospital	2024-03-15	12:00	15	52	L1:1, L2:5, L3:9
H002	Hvidovre Hospital	2024-03-15	16:00	19	71	L1:2, L2:7, L3:10
H003	Bispebjerg Hospital	2024-03-15	08:00	10	38	L1:1, L2:4, L3:5
H003	Bispebjerg Hospital	2024-03-15	12:00	14	58	L1:0, L2:6, L3:8
H003	Bispebjerg Hospital	2024-03-15	16:00	21	82	L1:2, L2:9, L3:10
H004	Amager Hospital	2024-03-15	08:00	7	29	L1:0, L2:2, L3:5
H004	Amager Hospital	2024-03-15	12:00	11	44	L1:1, L2:4, L3:6
H005	Frederiksberg Hospital	2024-03-15	08:00	9	41	L1:1, L2:3, L3:5
H005	Frederiksberg Hospital	2024-03-15	12:00	13	55	L1:0, L2:5, L3:8

✗ **Accuracy:** historical averages don't reflect current conditions; 4-hour averages hide minute-to-minute fluctuations users need

0 **Relevance:** wait times are relevant, but need current, not historical data

✗ **Representativeness:** historical data doesn't represent today's conditions

0 **Completeness:** historical completeness is ok, however, missing current data points

Average emergency department (ED) wait times collected from 5 major hospitals in the Capital Region over the past 5 years, complete, no errors.

How to think about data quality assessments?

The Five Facets of Data Quality Assessment

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ABSTRACT

Data-oriented applications, their users, and even the law require data of high quality. Research has divided the rather vague notion of data quality into various dimensions, such as accuracy, consistency, and reputation. To achieve the goal of high data quality, many tools and techniques exist to clean and otherwise improve data. Yet, systematic research on actually assessing data quality in its dimensions is largely absent, and with it, the ability to gauge the success of any data cleaning effort.

We propose five facets as ingredients to assess data quality: *data*, *source*, *system*, *task*, and *human*. Tapping each facet for data quality assessment poses its own challenges. We show how overcoming these challenges helps data quality assessment for those data quality dimensions mentioned in Europe's AI Act. Our work concludes with a proposal for a comprehensive data quality assessment framework.

1 The Many Dimensions of Data Quality

Data quality (DQ) has been an important research topic for the last decades [10, 43, 62], reflecting its critical role in all fields where data are used to gain insights and make decisions. A manifold of DQ dimensions exists that regard data and their properties from various perspectives and contribute to understanding and characterizing the complex nature of data [10, 62].

The high demand for DQ. Especially in the fast-moving landscape of *artificial intelligence* (AI), where data plays a pivotal role, the significance of DQ is dramatically increasing, so much so that literature calls this trend a paradigm shift from a model-centric view to a data-centric one [64]. Data-centric AI emphasizes the data and their impact on the underlying model [44, 45, 63]. Literature showed that DQ, with its various dimensions, significantly influences prediction accuracy [24, 36, 40, 45]. Domain-specific particulars provide a context that imposes specific requirements on DQ assessment, such as

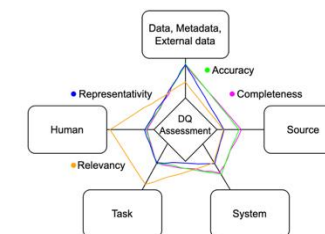


Figure 1: The five *facets* of DQ assessment and exemplary characteristics for DQ dimensions.

the *Health Insurance Portability and Accountability Act* (HIPAA), which focuses on privacy but promotes DQ dimensions, such as accuracy and completeness for ensuring trust [2].

Such requirements have also become part of regulation, as in the *General Data Protection Regulation* (GDPR) [25] and the *EU AI Act* [22]. For instance, the AI Act mentions in *Article 10* the DQ dimensions *representativity*, *accuracy* (free of errors), *completeness* and *relevancy* [22]. Similar initiatives to regulate DQ and AI are also being made by the United States [31] and China [52], which underlines the international interest in the topic of DQ.

Examining DQ is by no means just an academic problem [12]. Industry is also concerned about the impact of DQ on business [53]. Companies have shifted from internal “data gazing” [37] to hiring auditing firms for quality assurance. The literature shows that poor DQ has an enormous economic impact on organizations, either through loss of revenues or through additional internal costs [41, 50].

In addition to recognizing the relevance of DQ and understanding it in terms of the various dimensions, the goal is to improve DQ by cleaning the data. Yet, quality cannot be improved if it can-

5 facets of data quality

These facets can guide a more comprehensive assessment of the data quality

1. The data and metadata facet encompasses all the metrics that we'd traditionally list under technical aspects of "data quality"
2. We need to know how the data was collected and processed (*source*)
3. We need to be able to access and use it (*system*)
4. And 5. It has to make sense in the context of the (*task*) for the person (*human*) who will be working with it or using a system built on it.

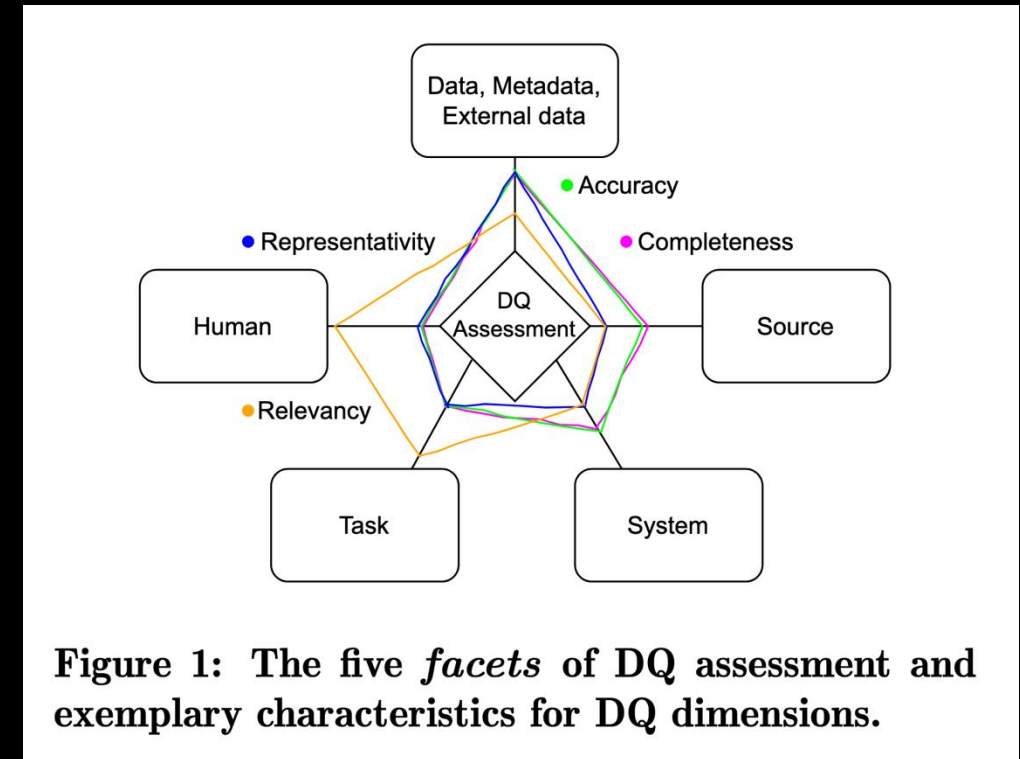


Figure 1: The five *facets* of DQ assessment and exemplary characteristics for DQ dimensions.

Data, Metadata

How to understand it

Qualities intrinsic to the data.

The data facet encompasses all challenges related to the data being assessed and its metadata,

Example dimensions

Accuracy describes the correspondence between a phenomenon in the world and its description as data

Representativity aims to ensure that the characteristics of the reference data are present in the considered data

Completeness refers to the extent to which data, including entities and attributes, are present according to the data schema

Source

How to understand it

This facet encompasses evaluating the data generation and collection processes, as well as assessing the source's integrity and organisational compliance (e.g., anonymisation).

Example dimensions

Provenance: information about the data origin, transformations, storage etc.

Reputation: The assessment requires evaluating a data source's credibility and reliability, and thus, considering historical reliability with data lineage.

Believability: The key challenge is to verify the data origin (time range), source transformations (data lineage), and involved entities (process).

System

How to understand it

The system facet concerns the physical aspects, including the infrastructure and technology for storing, handling, and accessing data, as well as technical compliance with legal requirements.

Example dimensions

Clarity includes documenting the system's architecture, data processing capabilities, interoperability with other systems, security features, and user interface aspects.

Auditability is crucial to verify compliance with regulations (like GDPR and security standards).

Recoverability: Assessing the ability to restore a prior state of the data requires knowledge about the file system, backup procedures (*clarity*) and long-term storage regulations (*auditability*).

Portability: The key challenge is to understand the storage system, including file formats (*clarity*) and interoperability standards (*auditability*).

Task

How to understand it

The task facet pertains to the specific use case and the context in which the data is employed. It is the context in which the data is going to be used and has to make sense. The task influences which parts of the data (e.g., columns, tuples) are considered and how well they represent the real world.

Example dimensions

Timeliness: The key challenge is defining an acceptable timeframe for tasks and to classify how data are considered up-to-date or relevant.

Relevance: The assessment involves balancing the need for complete information (relevance) against the risk of including unnecessary data that can violate legal requirements (risk).

Human

How to understand it

The human facet introduces a subjective view of stakeholders that interact with the data, perform the task, and interpret the results

Example dimensions

Ease of manipulation: Since manipulability can impact accessibility positively and data integrity negatively, the assessment must consider the users intent of manipulation.

Relevance: Determining relevant data varies by user perspective, experience, expertise etc. (intent). The evolving nature of relevancy with changing user needs, market trends, and legal standards complicates maintaining up-to-date assessments (survey design).

Exercise.

In pairs/groups, go to:

hdzajac.com -> Teaching Prototypes -> Data Quality Explorer

Inspect the synthetic dataset.

Assess five data quality facets?

Take some notes. 15 minutes.

We'll discuss them afterwards

Example, Data, Metadata

How to understand it

Qualities intrinsic to the data.

The data facet encompasses all challenges related to the data being assessed and its metadata,

Issues

Example, Data, Metadata

How to understand it

Qualities intrinsic to the data.

The data facet encompasses all challenges related to the data being assessed and its metadata,

Issues

No units (glucose, weight, HbA1c, BP)

Inconsistent dose formats (1000, 1000mg, 1g, 24 units)

3-month data gap (Feb-May 2024) - very noticeable!

Many missing values

Example, Source

How to understand it

This facet encompasses evaluating the data generation and collection processes, as well as assessing the source's integrity and organisational compliance (e.g., anonymisation).

Issues

Example, Source

How to understand it

This facet encompasses evaluating the data generation and collection processes, as well as assessing the source's integrity and organisational compliance (e.g., anonymisation).

Issues

PT001, PT002, PT003 have same-day readings from:

- Lab test (most reliable)
- Patient self-report
- Manual nurse entry

Example, System

How to understand it

The system facet concerns the physical aspects, including the infrastructure and technology for storing, handling, and accessing data, as well as technical compliance with legal requirements.

Issues

Example, System

How to understand it

The system facet concerns the physical aspects, including the infrastructure and technology for storing, handling, and accessing data, as well as technical compliance with legal requirements.

Issues

CPR numbers visible - NOT anonymised!

Data scattered across 4 locations

Archived data in .proprietary format (inaccessible)

Mix of .csv and .db formats

Task

How to understand it

The task facet pertains to the specific use case and the context in which the data is employed. It is the context in which the data is going to be used and has to make sense. The task influences which parts of the data (e.g., columns, tuples) are considered and how well they represent the real world.

Issues

RQ1 – What is the average HbA1c level in our diabetic patient population?

The Good:

The Bad:

Task

How to understand it

The task facet pertains to the specific use case and the context in which the data is employed. It is the context in which the data is going to be used and has to make sense. The task influences which parts of the data (e.g., columns, tuples) are considered and how well they represent the real world.

Issues

RQ1 – What is the average HbA1c level in our diabetic patient population?

The Good:

- HbA1c data is present – a few missing values
- Patients are Danish residents
- Appropriate outcome measure

The Bad:

- Selection bias: Where did these patients come from? One hospital? One clinic? One city?
- No sampling strategy documented: Are these representative of Danish diabetics?
- Missing data on patient demographics (age, gender, region, diabetes type, duration)

Task

How to understand it

The task facet pertains to the specific use case and the context in which the data is employed. It is the context in which the data is going to be used and has to make sense. The task influences which parts of the data (e.g., columns, tuples) are considered and how well they represent the real world.

Issues

RQ2 – How effective is Metformin at controlling blood glucose levels?

The Good:

The Bad:

Task

How to understand it

The task facet pertains to the specific use case and the context in which the data is employed. It is the context in which the data is going to be used and has to make sense. The task influences which parts of the data (e.g., columns, tuples) are considered and how well they represent the real world.

Issues

RQ2 – How effective is Metformin at controlling blood glucose levels?

The Good:

- Glucose measurements
- Metformin present in the medication column
- Multiple timepoints (can see change over time)

The Bad:

- No control group, everyone is on medication
- Can't separate the medication effect from natural variation or other interventions
- Patients on different medications (some on Metformin alone, others on combinations, PT002 on insulin)
- Dose varies (500mg vs 1000mg vs 1500mg)
- 3-month data gap

Task

How to understand it

The task facet pertains to the specific use case and the context in which the data is employed. It is the context in which the data is going to be used and has to make sense. The task influences which parts of the data (e.g., columns, tuples) are considered and how well they represent the real world.

Issues

RQ3 – What is patient satisfaction with their diabetes medication?

The Good:

The Bad:

Task

How to understand it

The task facet pertains to the specific use case and the context in which the data is employed. It is the context in which the data is going to be used and has to make sense. The task influences which parts of the data (e.g., columns, tuples) are considered and how well they represent the real world.

Issues

RQ3 – What is patient satisfaction with their diabetes medication?

The Good:

- Nothing

The Bad:

- Completely wrong data. Can't deduct patient satisfaction from these observations

Example, Human

How to understand it

The human facet introduces a subjective view of stakeholders that interact with the data, perform the task, and interpret the results

Issues

- Clinical researcher:
- Hospital admin:
- Medical doctor:

Example, Human

How to understand it

The human facet introduces a subjective view of stakeholders that interact with the data, perform the task, and interpret the results

Issues

- Clinical researcher:
 - Probably relevant data that they will be able to access and catch the issues
- Hospital admin:
 - Probably interested in RQ3, wrong data, and a difficult to assess format.
- Medical doctor:
 - May not realise data issues, e.g. lack of units, etc., because they may be implicitly understood thanks to domain knowledge.

Bias

Ordbog

Definitioner fra [Oxford Languages](#) · [Få flere oplysninger](#)

 **bias**
/ˈbiːəs/

noun

1. inclination or prejudice for or against one person or group, especially in a way considered to be unfair.

"there was evidence of **bias against** foreign applicants"

Lignende:

[prejudice](#)

[partiality](#)

[partisanship](#)

[favouritism](#)

[unfairness](#)



Bias

In the context of healthcare AI, bias can be defined as any systematic and/or unfair difference in how predictions are generated for different patient populations that could lead to disparate care delivery¹⁰. Through this, disparities related to benefit or harm are introduced or exacerbated for specific individuals or groups, eroding the capacity for healthcare to be delivered in a fair and equitable manner¹¹. The concept of "bias in, bias out", a derivative of

<https://doi.org/10.1038/s41746-025-01503-7>

Bias ☹️

Is it a property of the data or a property of how we use it?
For example,

A photograph of a neighbourhood isn't "biased" - it's just a photo of a neighbourhood. It becomes problematic if you claim it represents all neighbourhoods.

A census that only counts people who own properties isn't a census of the entirety of a population - it's a biased representation of the population that embeds 19th-century values about who matters.

Bias

Data is never “raw” or “objective”

Data is never “raw” or “objective”

Data are not simply ‘there’, waiting for us to be collected, stored and analysed, but instead data are always **imagined and created in particular historical, social, cultural and economical circumstances**. This draws us to the main theme and goal of the book, which is to reflect on the suggestion of Bowker that data are never ‘raw’ but always ‘cooked’. Precisely because data are constructed,... data ‘need to be understood as framed and framing, understood, that is, according to the uses to which they are and can be put’. The process of this framing requires further **investigations into the changing socio-economic contexts in which data are created and the heterogeneous social and cultural phenomena that are reduced to data**. In other words, we cannot take the data in Big Data for granted.

<https://doi.org/10.1080/1369118X.2014.920042>

Bias

Data is never “raw” or “objective”

date	location	crime_type	arrests_made
2024-01-15	Nordvest	theft	5
2024-01-16	Nordvest	assault	10
2024-01-17	Hellerup	theft	0

Bias

Data is never “raw” or “objective”

date	location	crime_type	arrests_made	interventions	convictions	police / 1000 people
2024-01-15	Nordvest	theft	5	6	1	2
2024-01-16	Nordvest	assault	10	12	2	2
2024-01-15	Hellerup	theft	0	4	0	0.25

Bias

Data is never “raw” or “objective”

date	location	crime_type	arrests_made	convictions	median_income	unemployment
2024-01-15	Nordvest	theft	5	1	30k	10%
2024-01-16	Nordvest	assault	10	2	30k	10%
2024-01-15	Hellerup	theft	0	0	50k	5%

date	location	crime_type	arrests_made	convictions	immigrant composition	non-western background
2024-01-15	Nordvest	theft	5	1	50%	35%
2024-01-16	Nordvest	assault	10	2	50%	35%
2024-01-15	Hellerup	theft	0	0	5%	2%

Bias

PATIENT ID	CPR NUMBER	DATE	GLUCOSE	BLOOD PRESSURE	WEIGHT	HBA1C	MEDICATION	DOSE	SOURCE	DATA LOCATION	ENTERED BY
PT001	*****_****	2024-01-15	145	128/82	78.5	7.2	Metformin	1000	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech
PT001	*****_****	2024-01-15	162	-	79	7.2	Metformin	1000mg	Patient Self-Report	/active/patient_portal_data.csv	Patient
PT001	*****_****	2024-01-15	140	130/84	78	7.2	Metformin	1000	Manual Entry	/active/nursing_vitals.db	RN Hansen
PT001	*****_****	2024-02-12	156	132/80	77.8	7.1	Metformin	1000	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech
PT001	*****_****	2024-02-12	168	-	-	-	Metformin	1g	Patient Self-Report	/active/patient_portal_data.csv	Patient
PT001	*****_****	2024-05-20	142	125/78	76.2	6.8	Metformin	1000	Lab	/active/lab_results_current.csv	Lab Tech
PT001	*****_****	2024-05-20	139	126/79	76	6.8	Metformin	1000mg	Manual Entry	/active/nursing_vitals.db	RN Olsen
PT002	*****_****	2024-01-08	198	145/95	92.3	8.9	Insulin Glargine	24	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech
PT002	*****_****	2024-01-08	203	-	93	-	Lantus	24 units	Patient Self-Report	/active/patient_portal_data.csv	Patient
PT002	*****_****	2024-01-22	189	142/92	91.8	8.7	Insulin Glargine	24	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech
PT002	*****_****	2024-02-05	182	138/88	90.5	8.5	Insulin Glargine	24	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech
PT002	*****_****	2024-02-05	195	-	-	-	Lantus	24u	Manual Entry	/active/nursing_vitals.db	RN Hansen
PT002	*****_****	2024-05-18	175	135/85	88.2	7.9	Insulin Glargine	28	Lab	/active/lab_results_current.csv	Lab Tech
PT002	*****_****	2024-06-02	171	132/82	87.5	7.8	Insulin Glargine	28	Lab	/active/lab_results_current.csv	Lab Tech
PT003	*****_****	2024-01-10	132	118/76	65.4	6.5	None	0	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech
PT003	*****_****	2024-01-10	128	120/78	65	-	None	-	Patient Self-Report	/active/patient_portal_data.csv	Patient
PT003	*****_****	2024-01-24	135	122/80	65.1	6.4	None	0	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech
PT003	*****_****	2024-02-14	138	125/82	64.8	6.6	Metformin	500	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech
PT003	*****_****	2024-02-14	142	-	-	-	Metformin	500mg	Manual Entry	/active/nursing_vitals.db	RN Olsen
PT003	*****_****	2024-05-22	129	120/77	64.2	6.3	Metformin	500	Lab	/active/lab_results_current.csv	Lab Tech
PT003	*****_****	2024-06-05	126	118/75	64	6.2	Metformin	500	Lab	/active/lab_results_current.csv	Lab Tech
PT004	*****_****	2024-01-12	167	135/88	85.2	7.8	Metformin	1500	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech
PT004	*****_****	2024-01-26	163	133/86	84.8	7.7	Metformin	1500	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech
PT004	*****_****	2024-02-09	159	130/84	84.2	7.5	Metformin	1500	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech
PT005	*****_****	2024-01-18	211	152/98	98.5	9.4	Metformin+Sitagliptin	1000+100	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech
PT005	*****_****	2024-02-01	205	148/95	97.8	9.2	Metformin+Sitagliptin	1000/100mg	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech
PT005	*****_****	2024-02-15	198	145/92	96.9	9.0	Metformin+Sitagliptin	1000+100	Lab	/archive/2024/lab_results_Q1.proprietary	Lab Tech

Bias

Bias in our test data:

1. Diabetes defined only as a relationship of glucose, medication, weight, blood pressure.

However...

Associations of SES With Diabetes Incidence, Prevalence, and Outcomes

Income, education, and occupation show a graded association with diabetes prevalence and complications across all levels of SES, up to the very top. Those lower on the SES ladder are more likely to develop T2DM, experience more complications, and die sooner than those higher up on the SES ladder (46,47). The higher a person's income, the greater their educational attainment, and the higher their occupational grade, the less likely they are to develop T2DM or to experience its complications. The gradient is steeper at the bottom, however, and research has focused primarily on those with the lowest levels of income and education.

<https://doi.org/10.2337/dci20-0053>

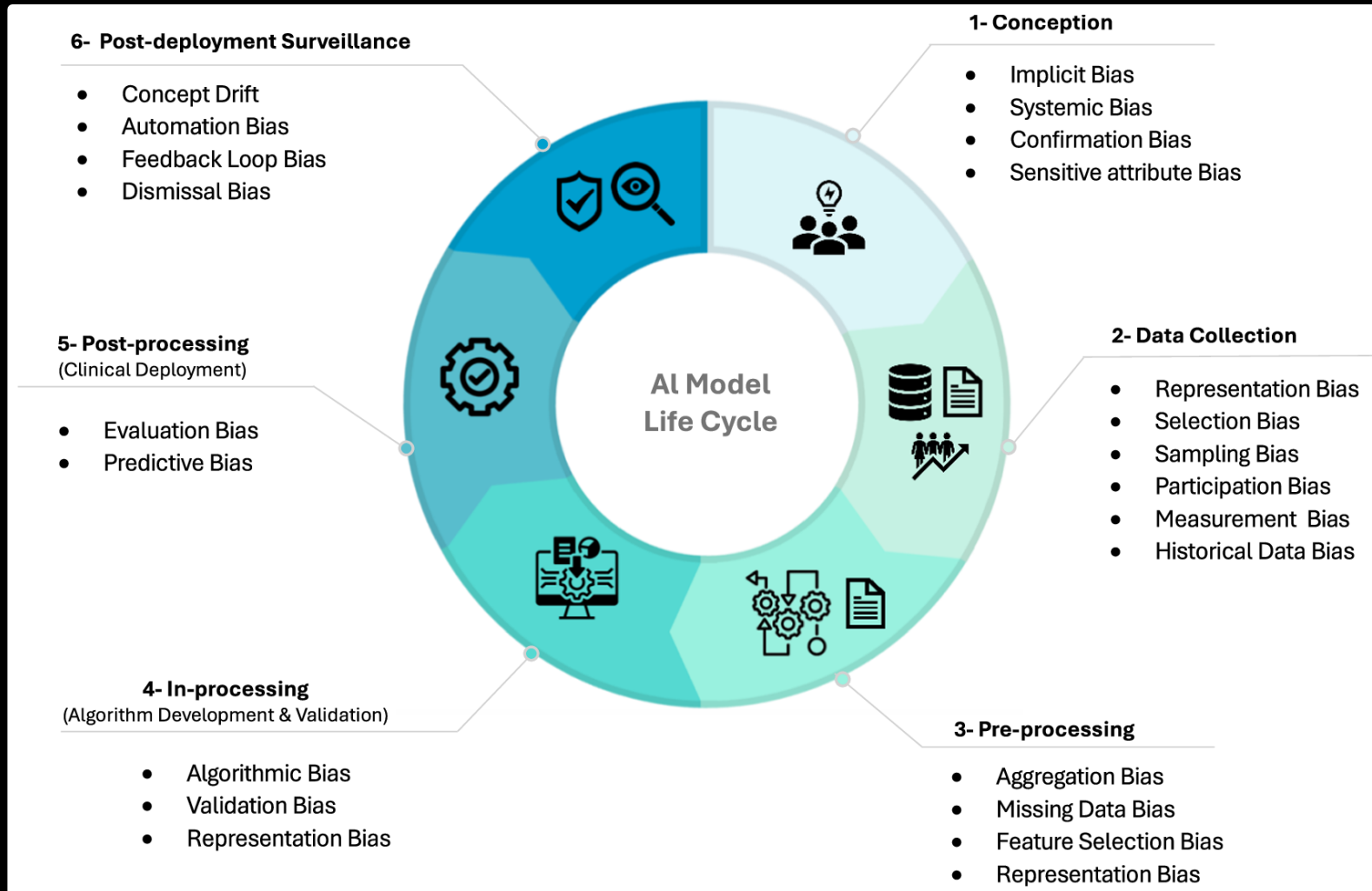
You cannot see that in our data

Bias

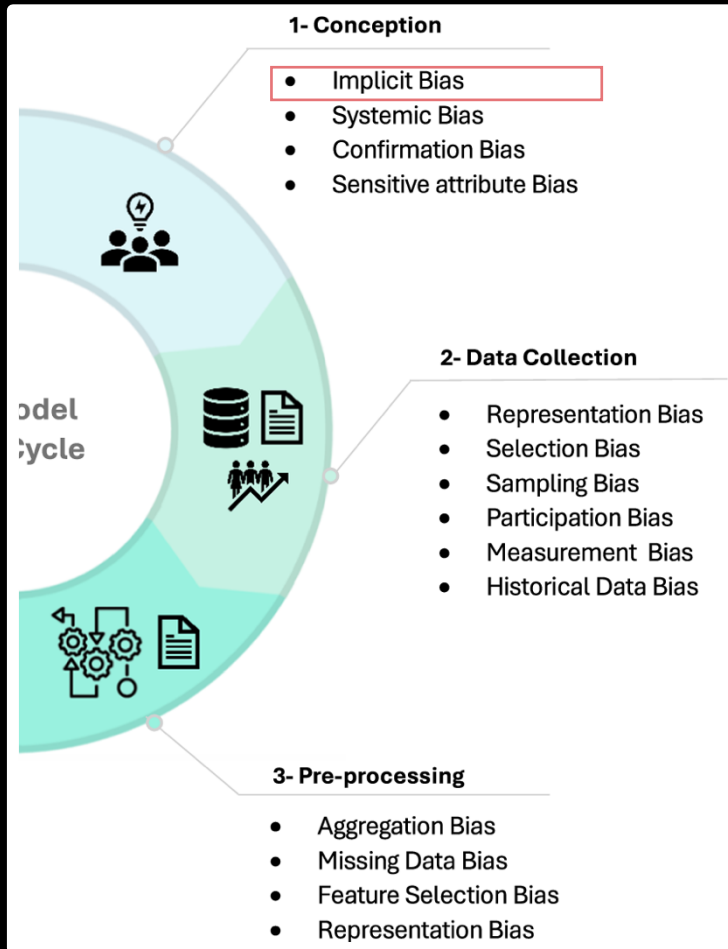
Bias in our test data:

1. Diabetes defined only as a relationship of glucose, medication, weight, blood pressure.
2. Data about effectiveness of the treatment does not account for
 - Income inequality influencing the socioeconomic determinants of diabetes
 - Non-Danish speaker adherence and feeling of support
 - Demographic data: age, ethnic background
 - How these factors affect the treatment?
3. Very little sample size

Bias

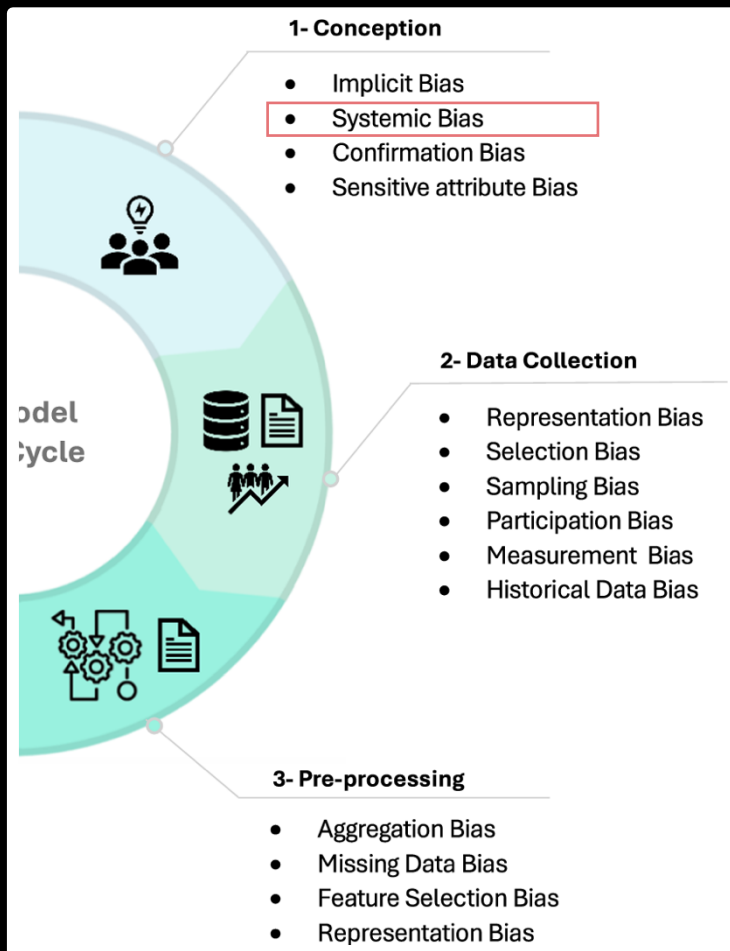


Bias



Implicit bias occurs when subconscious attitudes or stereotypes about a person's or group's characteristics, such as birth sex, gender identity, race, ethnicity, age, and socioeconomic status, become embedded in how individuals behave or make decisions.

Bias

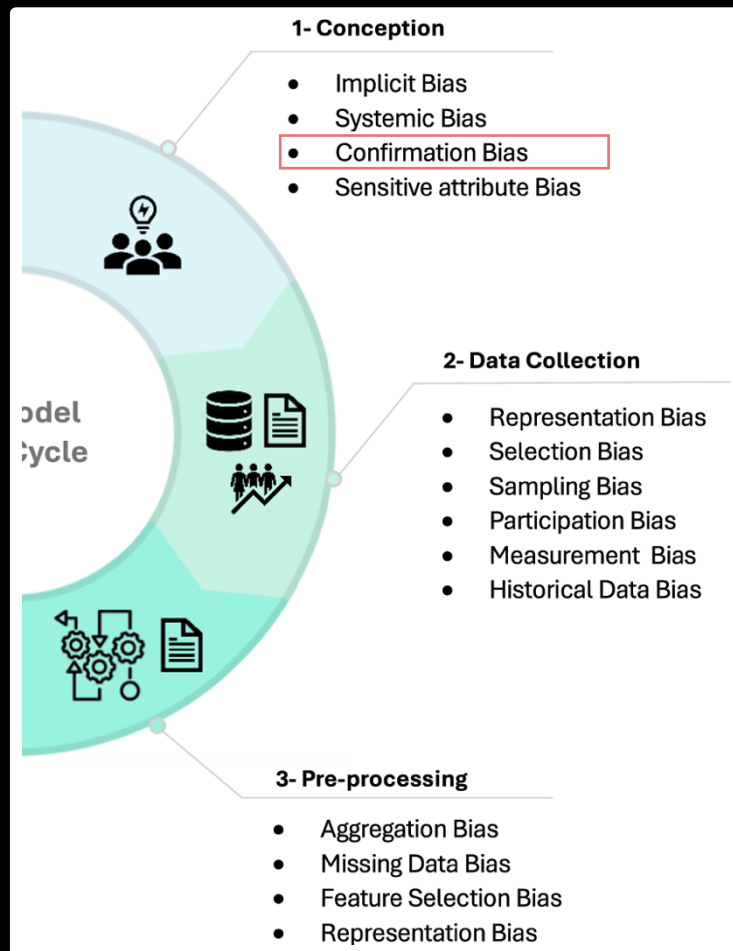


Implicit bias occurs when subconscious attitudes or stereotypes about a person's or group's characteristics, such as birth sex, gender identity, race, ethnicity, age, and socioeconomic status, become embedded in how individuals behave or make decisions.

While related to implicit bias, **systemic bias** extends to encompass broader institutional norms, practices, or policies that can lead to societal harm or inequities. The origins of systemic bias are more structural in nature and act at higher societal levels than implicit bias.

For example, policies – funding education from local property taxes. Visible in the US. Rich white neighbourhoods have better schools. Racism under the pretext of zoning laws. Bail decisions in the US justice system.

Bias



<https://www.nature.com/articles/s41746-025-01503-7>

Implicit bias occurs when subconscious attitudes or stereotypes about a person's or group's characteristics, such as birth sex, gender identity, race, ethnicity, age, and socioeconomic status, become embedded in how individuals behave or make decisions.

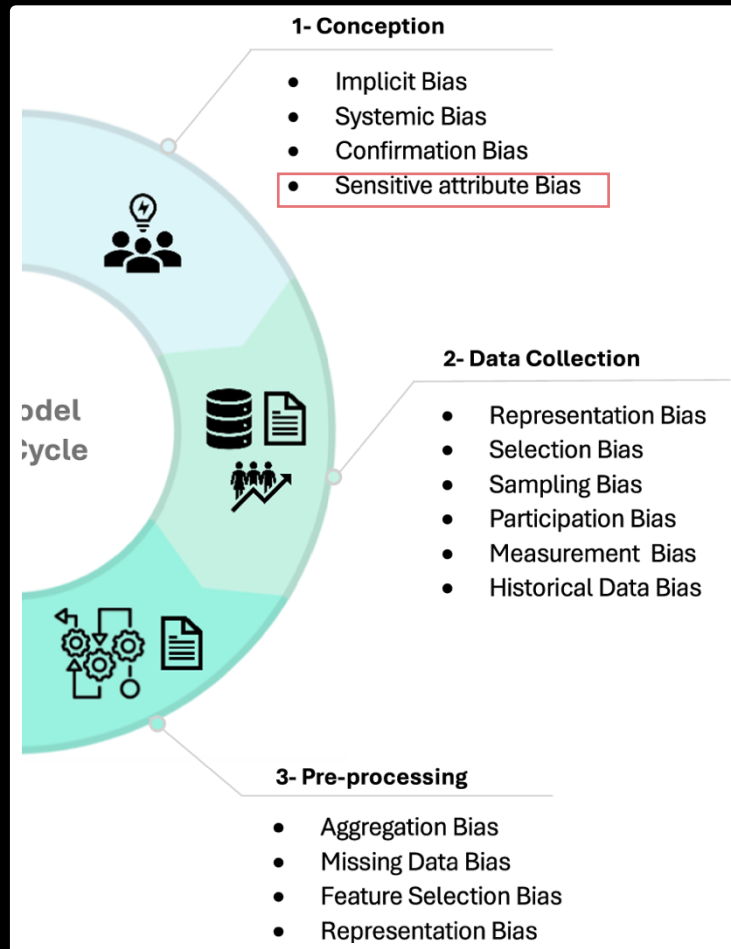
While related to implicit bias, **systemic bias** extends to encompass broader institutional norms, practices, or policies that can lead to societal harm or inequities. The origins of systemic bias are more structural in nature and act at higher societal levels than implicit bias.

During model development or data creation, people may consciously or subconsciously select, interpret, or give more weight to data that confirms their beliefs, overemphasising certain patterns while ignoring other patterns that don't fit expectations. This form of human-mediated bias is called **confirmation bias**.

date	location	crime_type	arrests_made	convictions	median_income	unemployment
2024-01-15	Nordvest	theft	5	1	30k	10%
2024-01-16	Nordvest	assault	10	2	30k	10%
2024-01-15	Hellerup	theft	0	0	50k	5%

date	location	crime_type	arrests_made	convictions	immigrant composition	non-western background
2024-01-15	Nordvest	theft	5	1	50%	35%
2024-01-16	Nordvest	assault	10	2	50%	35%
2024-01-15	Hellerup	theft	0	0	5%	2%

Bias



<https://www.nature.com/articles/s41746-025-01503-7>

Implicit bias occurs when subconscious attitudes or stereotypes about a person's or group's characteristics, such as birth sex, gender identity, race, ethnicity, age, and socioeconomic status, become embedded in how individuals behave or make decisions.

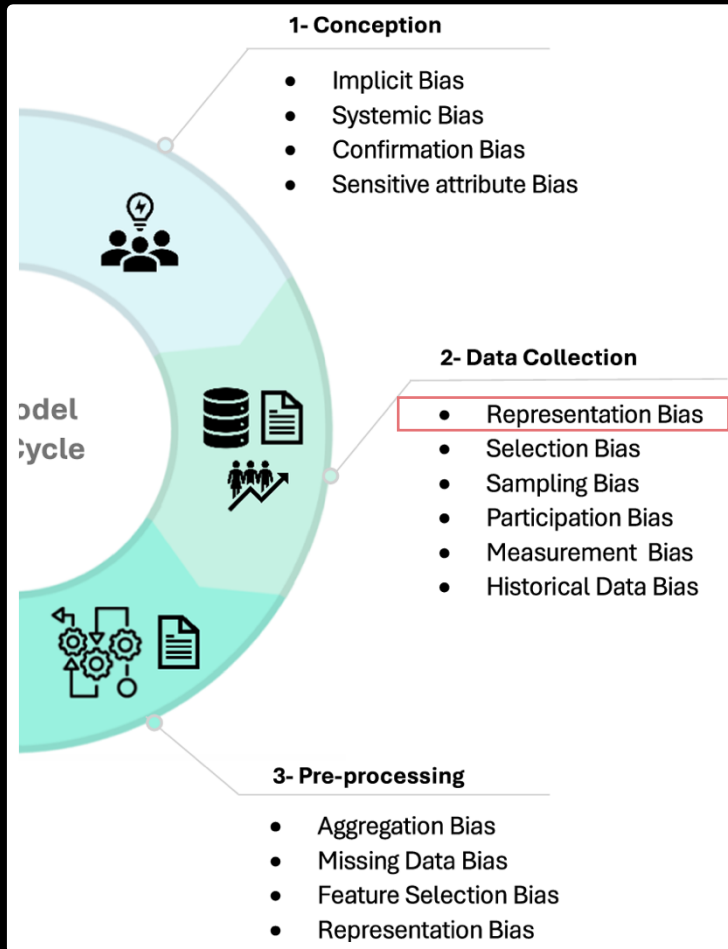
While related to implicit bias, **systemic bias** extends to encompass broader institutional norms, practices, or policies that can lead to societal harm or inequities. The origins of systemic bias are more structural in nature and act at higher societal levels than implicit bias.

During model development or data creation, people may consciously or subconsciously select, interpret, or give more weight to data that confirms their beliefs, overemphasising certain patterns while ignoring other patterns that don't fit expectations. This form of human-mediated bias is called **confirmation bias**.

Sensitive attribute biases, such as age, gender, or ethnicity, may be introduced to "proxy variables" (i.e. surrogates) for more complex features not adequately described by training data to enhance the accuracy and equity of healthcare AI models. However, their use is complex and can lead to the propagation of the bias they aimed to reduce.

For example, a study by Obermeyer et al. found that an algorithm designed to identify high-risk patients based on predicted healthcare costs exhibited racial bias, underestimating the needs of Black patients. This bias stemmed from using healthcare costs as a proxy for illness severity; systemic barriers like reduced healthcare access, financial constraints, and lower trust levels led to lower costs for Black patients, causing the algorithm to misjudge their risk.

Bias



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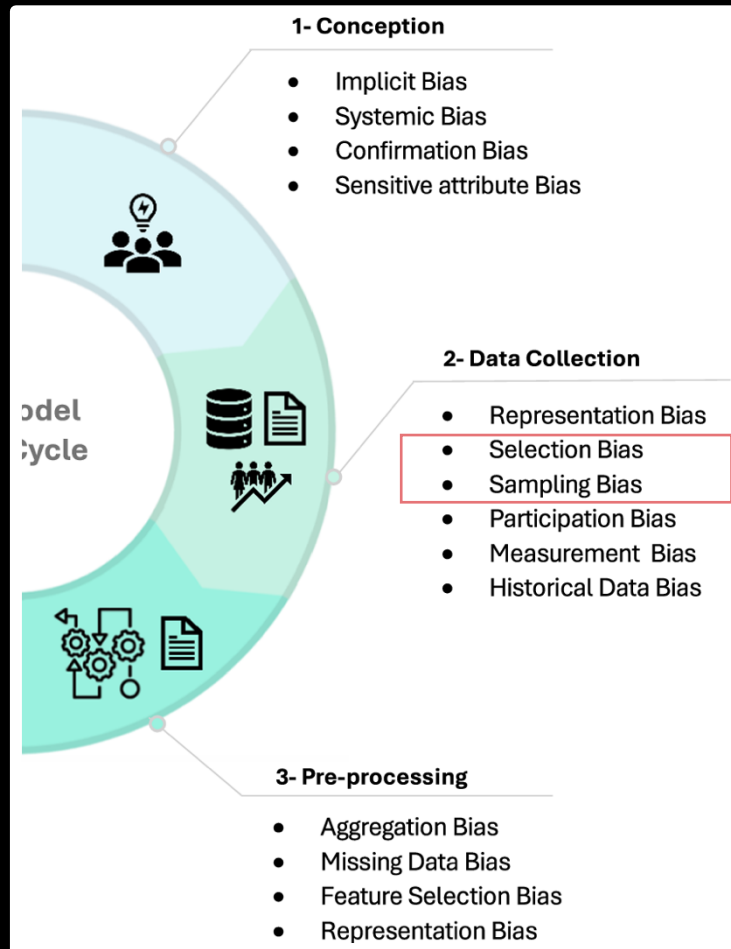
Representation bias refers to insufficient diversity in training data and is a major source of bias that limits the generalizability of healthcare AI models to unique environments or populations.

Home Office admits facial recognition tech issue with black and Asian subjects

Calls for review after technology found to return more false positives for 'some demographic groups' on certain settings

● [UK politics live - latest updates](#)

Bias

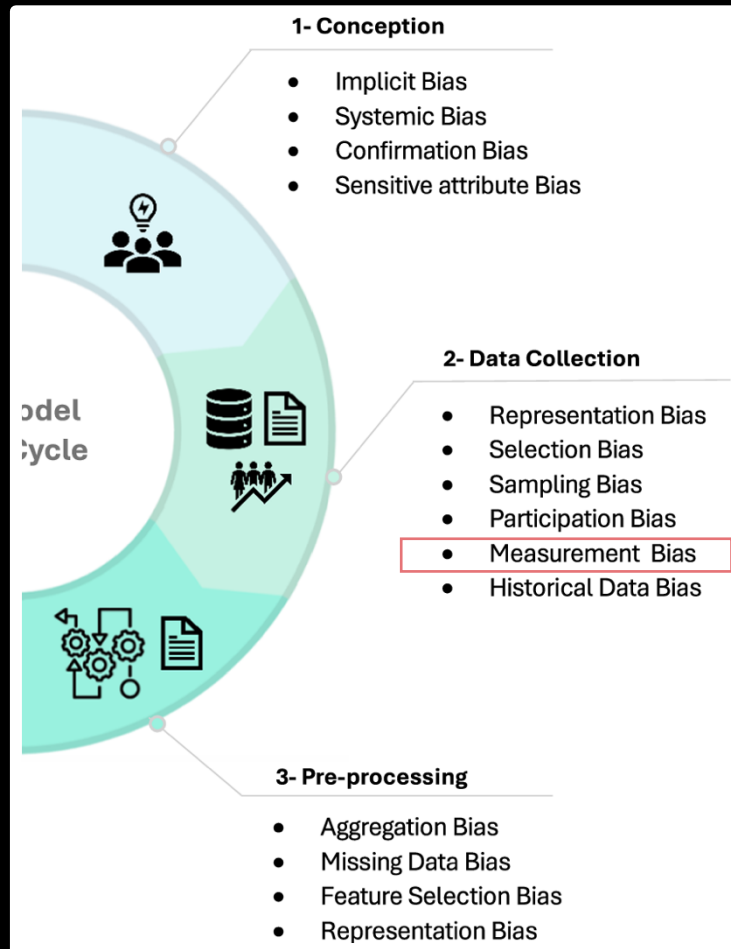


Representation bias refers to insufficient diversity in training data and is a major source of bias that limits the generalizability of healthcare AI models to unique environments or populations.

Representation bias can also arise from selection or sampling bias. **Selection bias** occurs when the way data is chosen or collected inadvertently favours certain groups or characteristics, leading to a non-representative sample. **Sampling bias** is a form of selection bias that arises from non-random sampling of subgroups, creating data patterns that are not generalizable to new populations. **Participation bias** or self-selection bias are also strong contributors to representation bias in research-generated datasets.

For example, patients who are sicker, have higher comorbidities, or are unable to participate in longitudinal research protocols for geographic or economic reasons may not be represented.

Bias



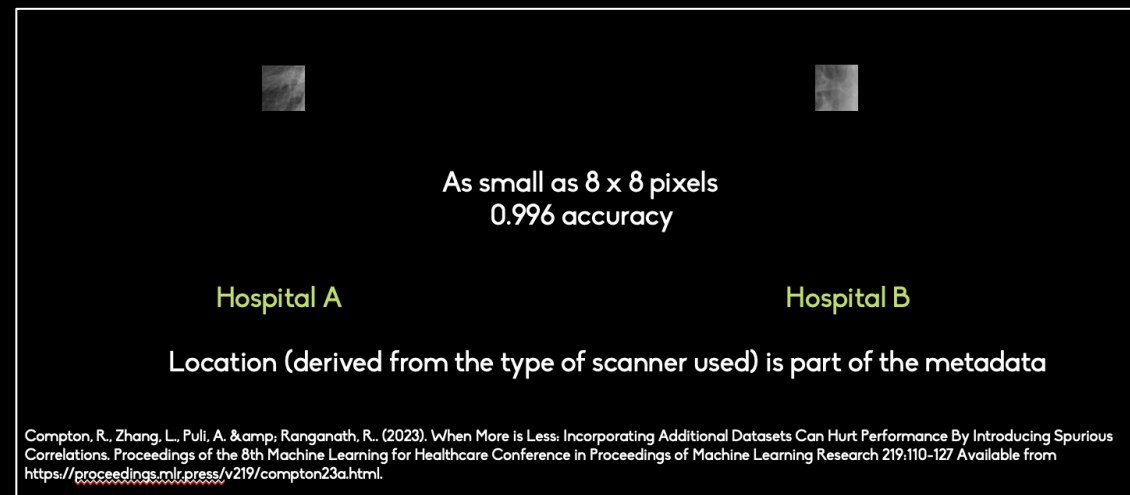
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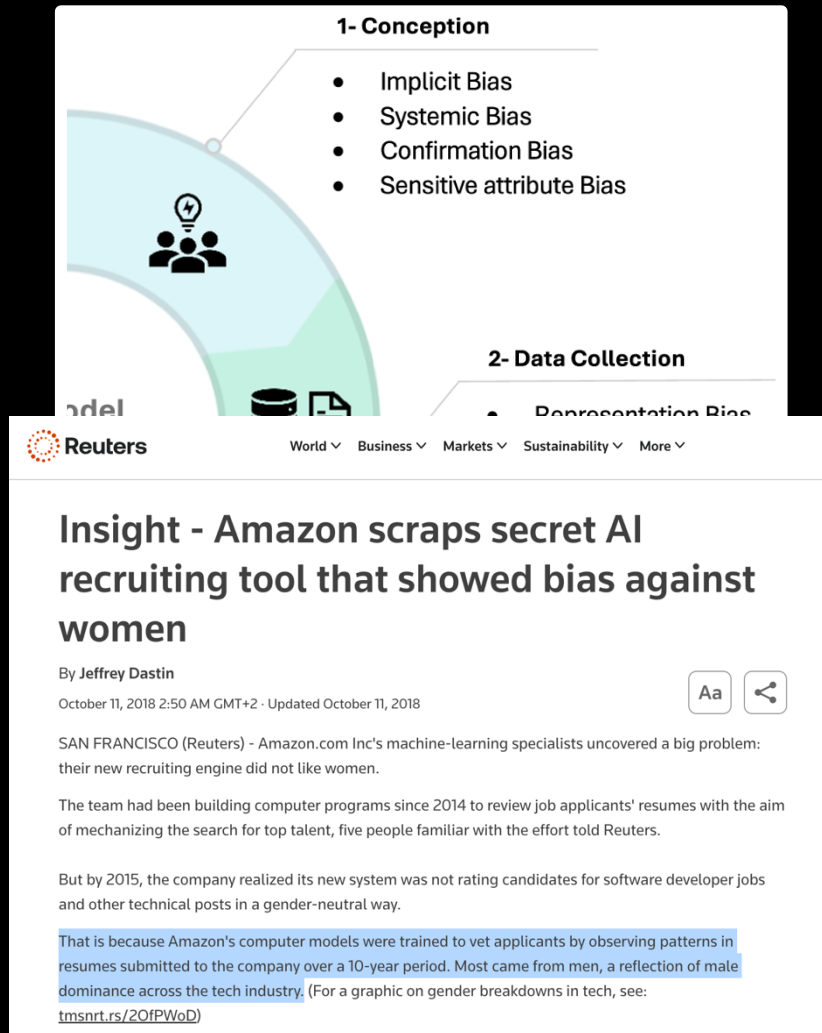
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Participation bias or self-selection bias are also strong contributors to representation bias in research-generated datasets.

Large-scale data resources required for AI model development and validation are often sourced from multiple hospital sites, each with unique methods of data acquisition. Such variations are often not related to biological factors, rather are related to differences in data acquisition or processing. These **measurement biases** result in systematic differences that alter the representation of variables.



Bias



1- Conception

- Implicit Bias
- Systemic Bias
- Confirmation Bias
- Sensitive attribute Bias

2- Data Collection

Representation Bias

Insight - Amazon scraps secret AI recruiting tool that showed bias against women

By Jeffrey Dastin

October 11, 2018 2:50 AM GMT+2 · Updated October 11, 2018

SAN FRANCISCO (Reuters) - Amazon.com Inc's machine-learning specialists uncovered a big problem: their new recruiting engine did not like women.

The team had been building computer programs since 2014 to review job applicants' resumes with the aim of mechanizing the search for top talent, five people familiar with the effort told Reuters.

But by 2015, the company realized its new system was not rating candidates for software developer jobs and other technical posts in a gender-neutral way.

That is because Amazon's computer models were trained to vet applicants by observing patterns in resumes submitted to the company over a 10-year period. Most came from men, a reflection of male dominance across the tech industry. (For a graphic on gender breakdowns in tech, see: tmsnrt.rs/Z0IPWoD)

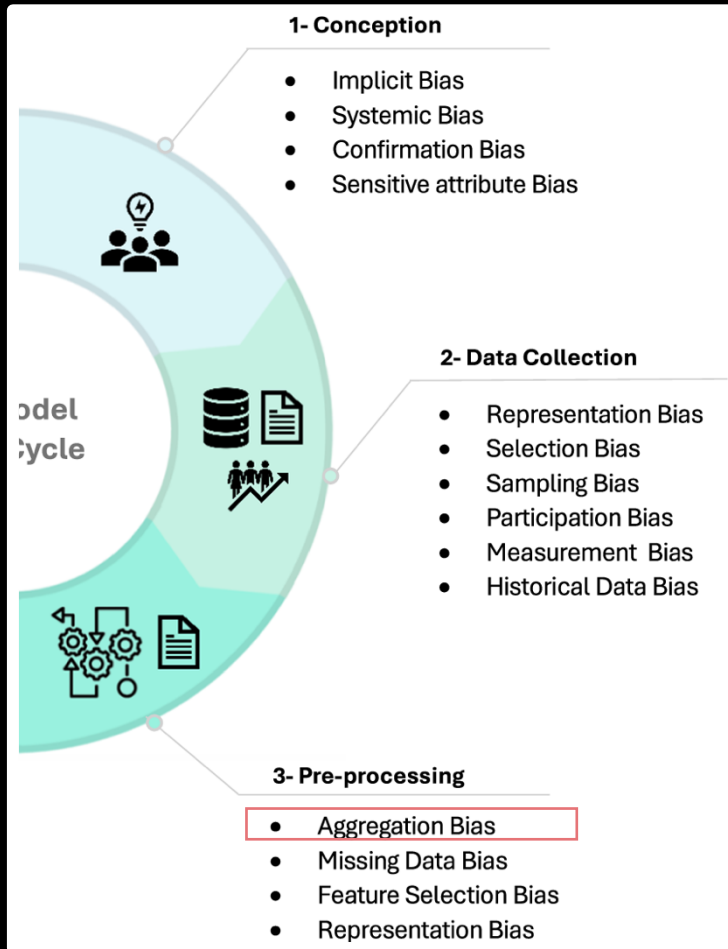
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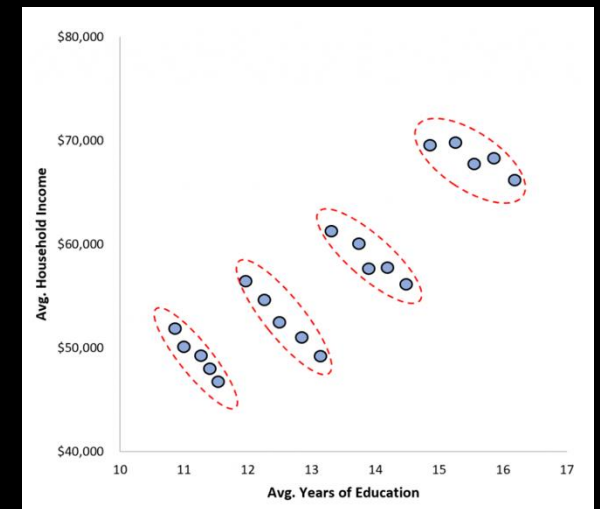
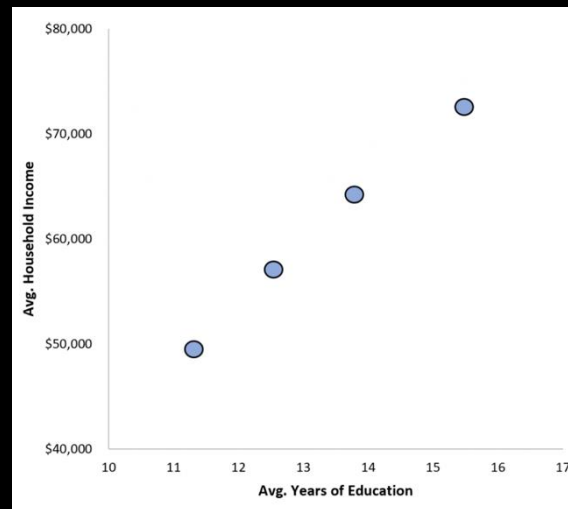
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Effort must be undertaken to ensure awareness of historical data that contains outdated practices, underrepresented groups, or past healthcare disparities not representing current real-world scenarios. **(Historical data bias)**

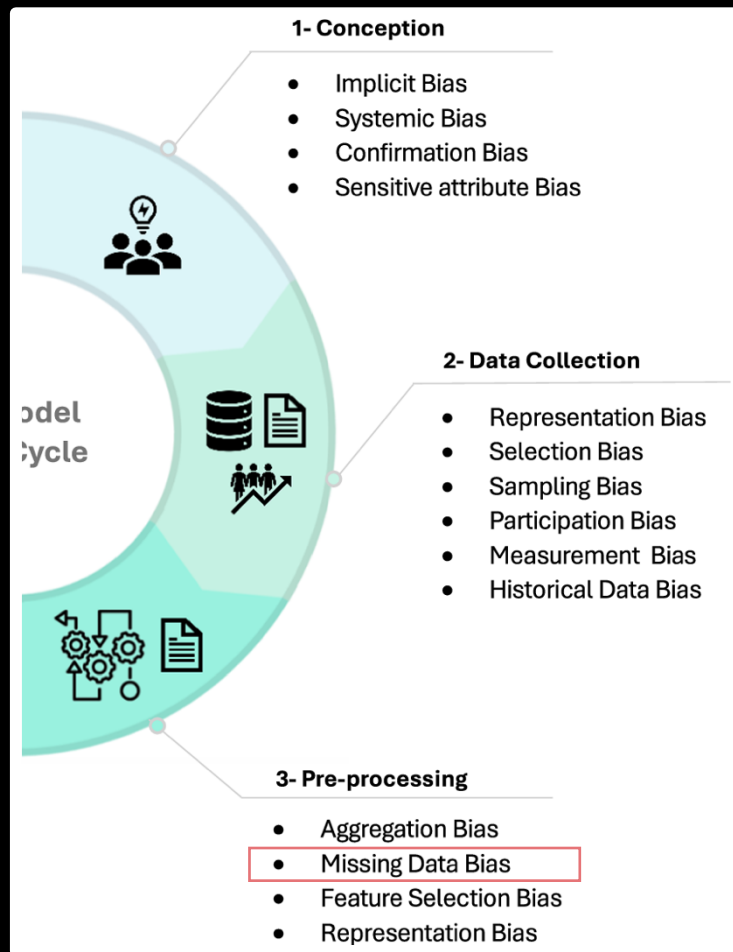
Bias



Aggregation bias occurs when it is wrongly assumed that the trends seen in aggregated data also apply to individual data points. While aiming to reduce aggregation bias, shifts toward disaggregated data can produce overly granular datasets prone to noise, reducing model generalizability.



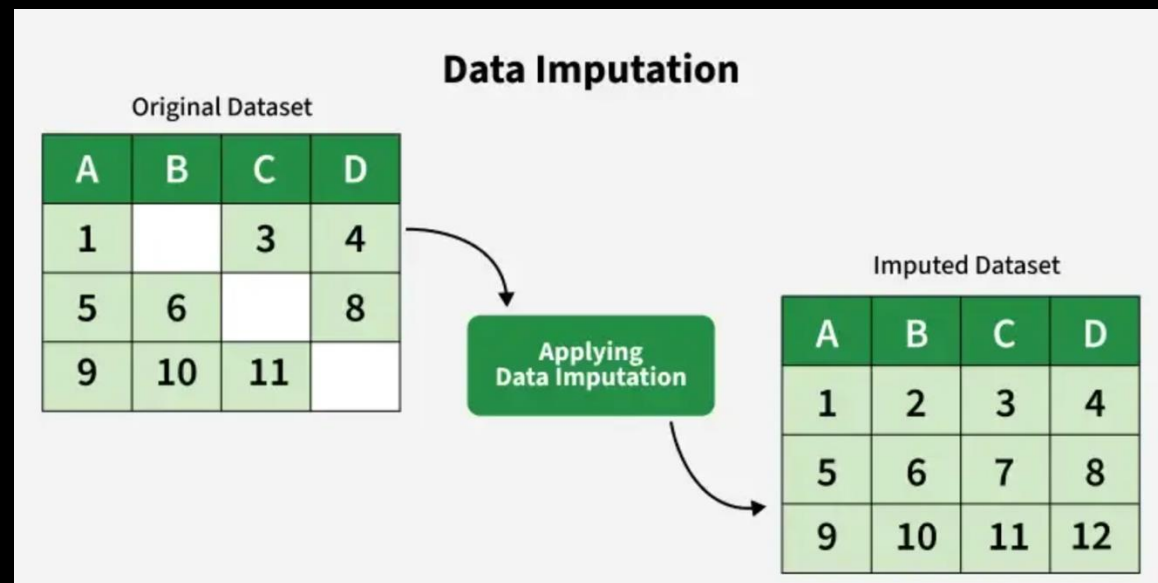
Bias



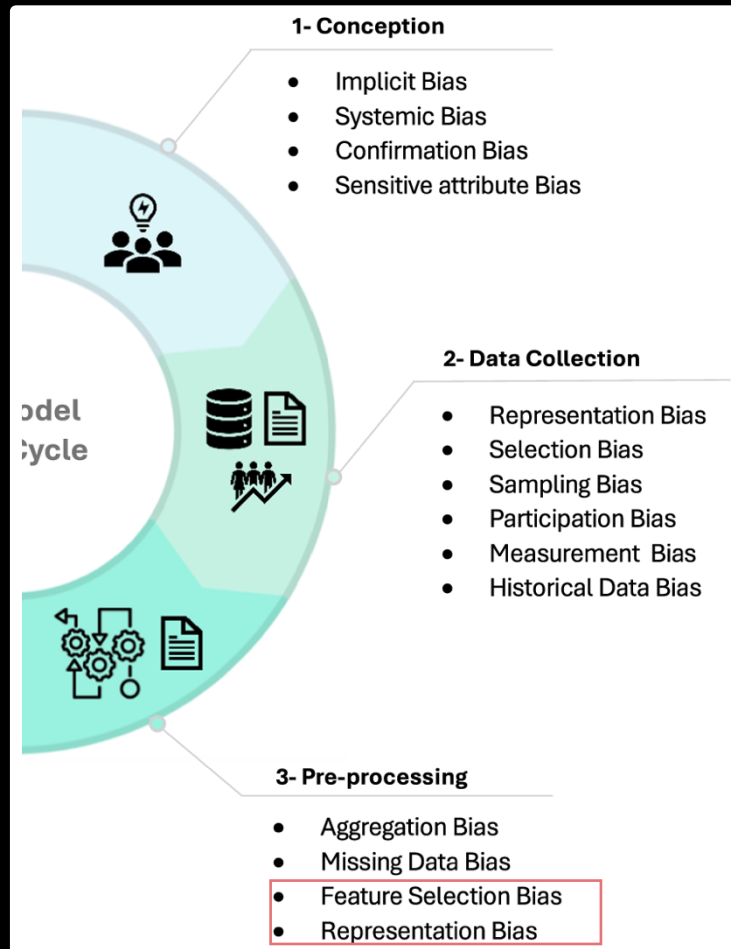
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The more data is missing, the more biased the estimations are. When handling missing data bias, multiple imputation may introduce inaccuracies, particularly for non-random missingness or substantial data gaps. Data augmentation to address sparsity can generate synthetic data that fails to appropriately reflect true diversity, thus reinforcing biases.



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Feature selection bias is the selective inclusion or exclusion of variables during model development based on preconceived ideas or beliefs about the planned task. Extending beyond human confirmation bias, the forced inclusion of variables hypothesised to influence performance or deemed of high priority, commonly referred to as Sensitive Variables, is a common practice foundationally motivated to avoid bias. Variables, such as age, sex, and race, are often deemed “sensitive” because they represent personal characteristics that predictive algorithms should ideally not discriminate against.

Bias

There are many ways to mitigate biases.

The best way to start is to be aware that they are there.

Conception	• Implicit Bias: Provide training on recognizing and mitigating biases for everyone involved in AI development. Include individuals from varied backgrounds. Explore your unconscious biases and commit to regular review and training sessions. • Systemic Bias: Analyze and understand the broader context of data and AI application areas. Review organizational policies and practices for unrecognized biases. Recognize the impact of historical discrimination to better perceive current biases. • Confirmation Bias: Encourage a culture of critical thinking and questioning assumptions among the team. Utilize multiple perspectives in model design. • Sensitive attribute Bias: Stay mindful of unconscious assumptions about age, birth sex, gender, ethnicity, race, sexual orientation, religion in the research question. (These biases may happen in all phases)
Data Collection	• Representation Bias: Collect data across diverse geographic locations, healthcare settings, and patient demographics. Actively include underrepresented groups. Leverage accessible data sources (such as Open Science Practices and STANDING together). Understand origins of this bias (selection, sampling, participation biases or historical data). • Selection Bias: Use stratified sampling for diversity. Apply blinding to prevent bias, and pre-register studies for transparency. • Sampling Bias: Match sampling frame with target population, simplify surveys, follow up with non-responders, avoid convenience sampling, and use random sampling. • Participation Bias (or self-selection bias): Offer incentives to encourage diverse participation. Use multiple survey modes: online, paper, phone. Lower barriers to entry. Use multiple channels to reach a wider audience. • Measurement Bias: Improve the design and calibration of the measurement system, standardize data collection, and apply statistical bias adjustments.
Pre-processing	• Aggregation Bias: Utilize disaggregated data, apply regression analysis, perform sensitivity analysis, implement hierarchical modeling, and adopt alternative measures. The optimal approach will depend on the specific data and research question. • Missing Data Bias: Maximize data collection efforts, apply multiple imputation techniques, augment existing data, follow up with participants, and conduct valid statistical analyses. • Feature selection Bias: Select features based on their relevance and potential impact on model outcomes, carefully avoiding assumptions based on stereotypes or incomplete information. Be critical of using proxy variables. • Representation bias: Employ data augmentation techniques, such as generating synthetic data or using data interpolation, and apply fairness-aware data pre-processing techniques, such as re-weighting or re-sampling, to balance the dataset.

Interoperability

An ability of two or more systems or components to exchange information and to use the information that has been exchanged.

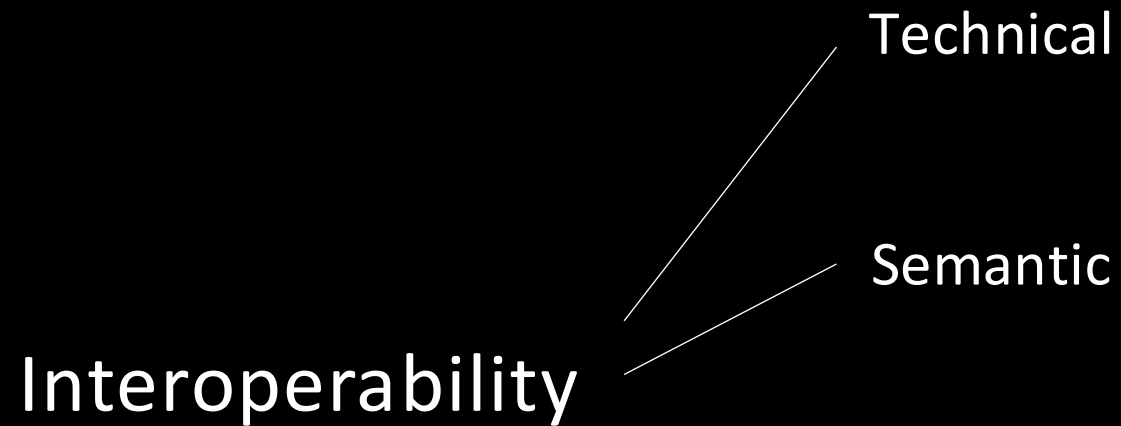
Interoperability

Interoperability

Technical

Technical interoperability moves data from system A to system B, neutralising the effects of distance. Technical interoperability is domain-independent. It does not know or care about the meaning of what is exchanged. It's the technology layer.

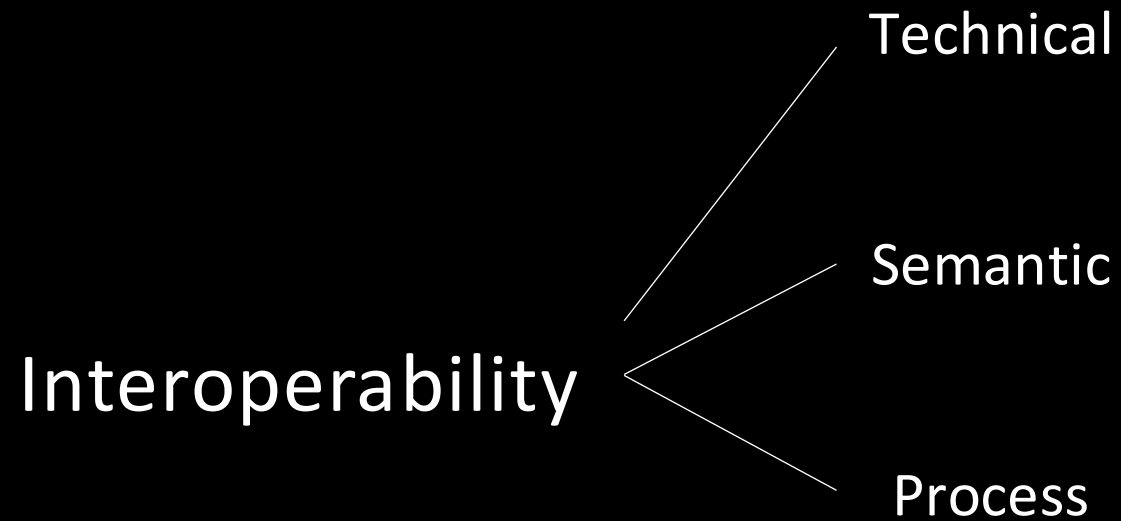
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Semantic interoperability allows computers to share, understand, interpret and use data without ambiguity. Semantic interoperability is specific to domain and context and requires the use of unambiguous codes and identifiers. This is the data layer.

Interoperability

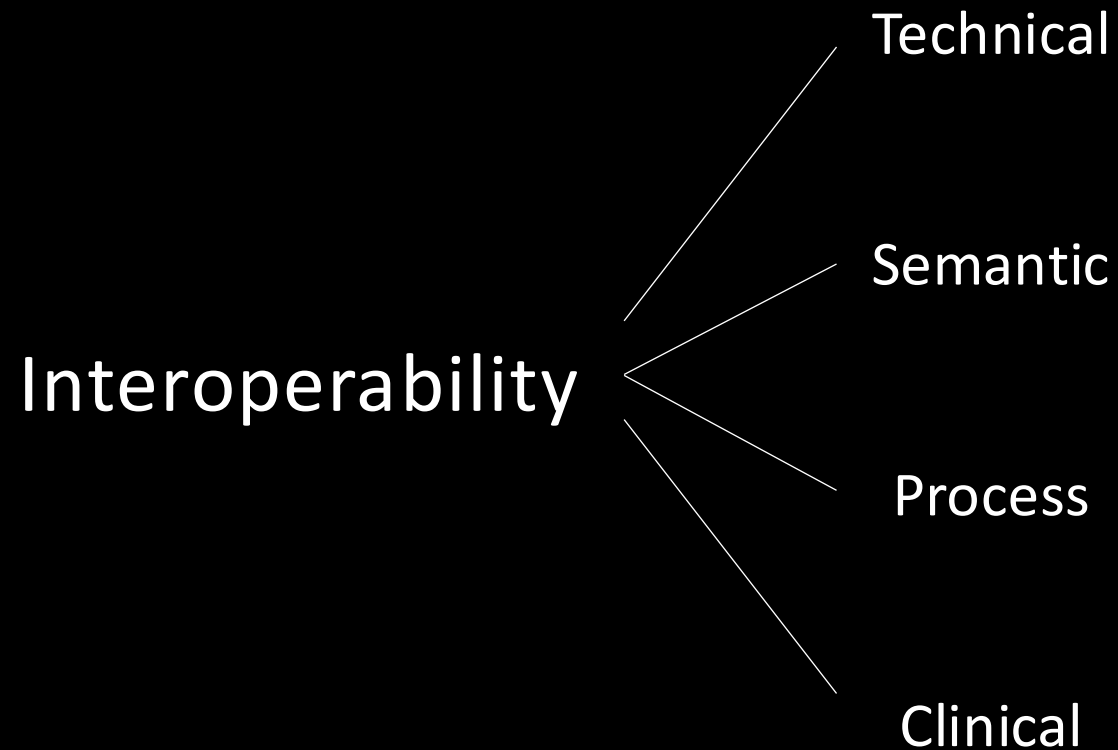


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Process interoperability is achieved when people share a common understanding across a network, business systems interoperate, and work processes are coordinated. People benefit when they use information from elsewhere in their day-to-day work.

Interoperability



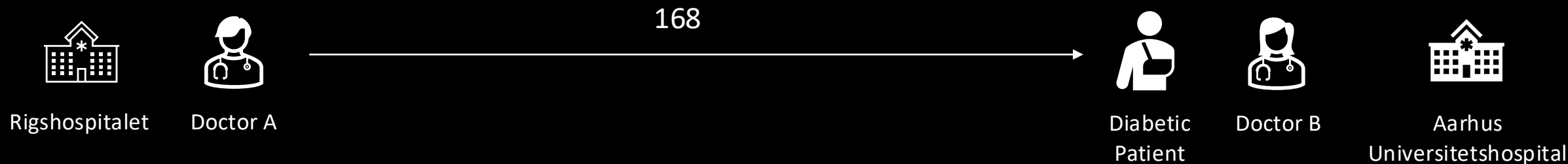
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Clinical interoperability is the ability for two or more clinicians in different care teams to transfer patients and provide seamless care to the patient. On its own, exchanging data achieves nothing. Only when people use new information in some way that differs from what they would have done without it, can we obtain different results and outcomes. In healthcare clinical interoperability is what matters.

Interoperability example: Communicating glucose results



Patient: 010185-2847 (Male, 39 years old)

Diagnosis:

Type 2 Diabetes Mellitus (newly diagnosed December 2025)

Clinical History:

Dec 1: Initial diagnosis, fasting glucose 180 mg/dL, HbA1c 7.2%

Dec 1: Started pharmacological treatment

Dec 8: Fasting glucose 172 mg/dL (improving)

Dec 15: Fasting glucose 168 mg/dL
(THIS RESULT - continuing to improve)

70–99 mg/dL – normal

100–125 mg/dL – prediabetes

> 125 mg/dL - diabetes

> 250 mg/dL – acute emergency - hyperglycemia

The patient had to go to the ED in Aarhus :(

In Aarhus, they cannot check the patient's blood sugar levels, but they can get the latest results from Rigshospitalet.

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(Communicate this result - continuing to improve)

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Discuss in pairs/groups the four interoperability aspects.

- How could this value be sent?
- How to make it unambiguous?
- How was this data obtained? Can it be shared? How will it reach Doctor B?
- Is this value helpful to Doctor B?
- What other information may be necessary?
- What challenges do you foresee?

15 min

Exercise: communicating glucose results

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Exercise: communicating glucose results

```
{
  "header": {
    "messageType": "HL7_FHIR_R4",
    "messageId": "MSG-2025-12-15-001",
    "transmissionProtocol": "HTTPS",
    "apiEndpoint": https://aarhus-ed.dk/api/observations

    "system_source": "Rigshospitalet-EHR",
    "intended_recipient": "Aarhus-ED-Workflow",
    "priority": "stat" },
  "payload":{
    "patient_id": "010185-2847",
    "clinical_context": {
      "diagnosis": {
        "code": "ICD10 E11.9",
        "text": "Type 2 Diabetes Mellitus",
        "diagnosis_date": "2025-12-01T10:00:00Z" },
      "current_medications": [ {
        "name": "Metformin",
        "dose": "1000mg",
        "frequency": "BID",
        "start_date": " 2025-12-01T10:00:00Z ",
        "last_dose": "2026-01-15T20:00:00Z" } ],},
    "observation": {
      "test_type": {
        "code": "LOINC 14771-0",
        "display": "Glucose [Mass/volume] in Blood by Fasting" },
      "result": {
        "value": 168,
        "unit": "mg/dL" },
      "timestamp": "2025-12-15T08:00:00Z",
    }
  }
}
```

Technical

JSON format. Structured data. UTF-8 encoding. A secure connection between the hospitals using API.

Semantic

LOINC code (14771-0) - international standard, explicit unit (mg/dL), timestamp in ISO 8601, clear field names, distinguishes "fasting glucose" (not just "glucose")

Process

Priority flag suggests the handling of the information, source and recipient specification aims to allow for workflow integration.

Clinical

Information about the diagnosis and the treatment allows the recipient to contextualise and make sense of the 168 value.

Technical interoperability challenges

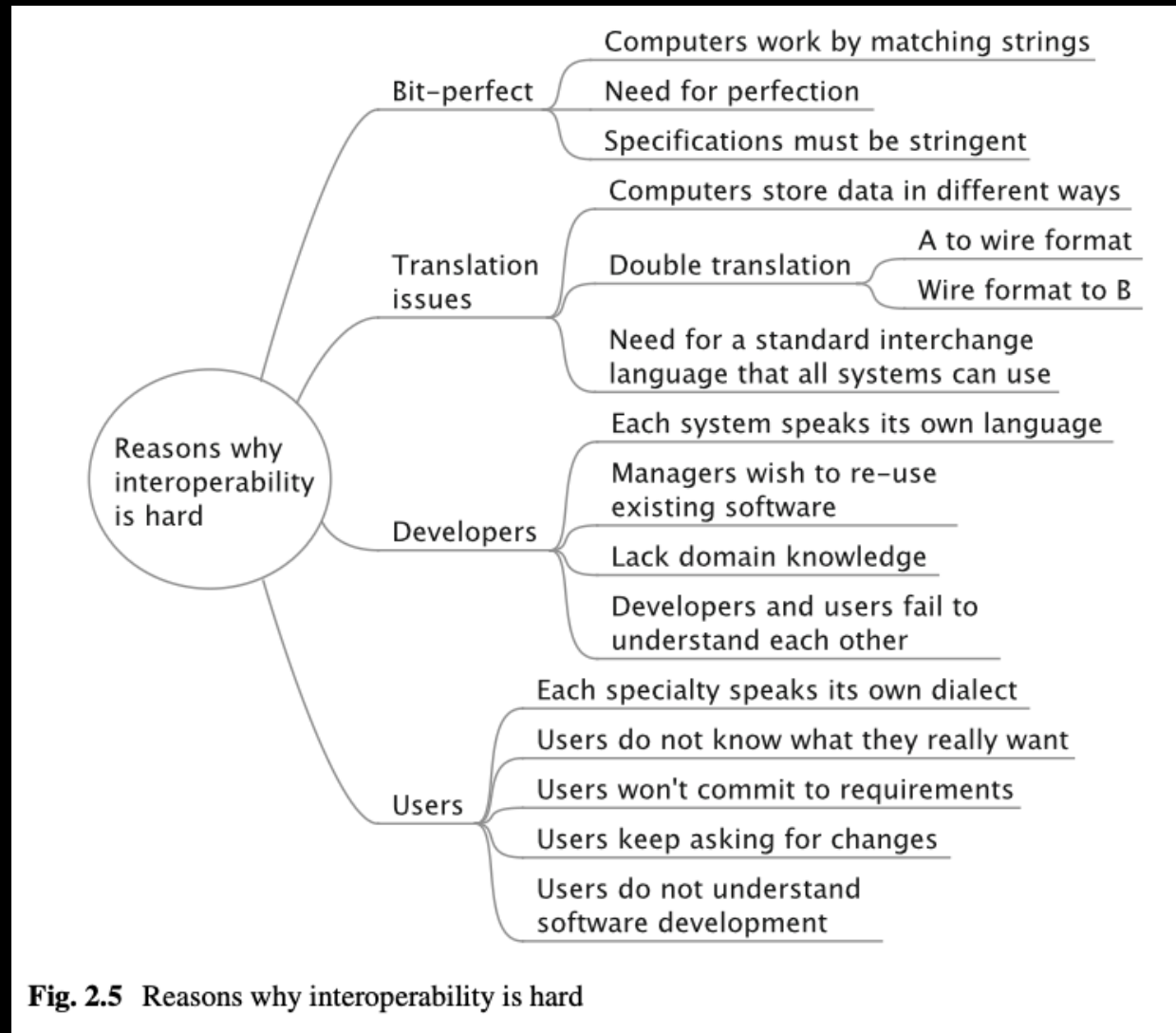


Fig. 2.5 Reasons why interoperability is hard

Socioeconomic challenges

Differences in clinical practice

Differences in clinical terminology

Governance and maintenance issues

Legal & regulatory challenges – GDPR etc.

Technical interoperability challenges

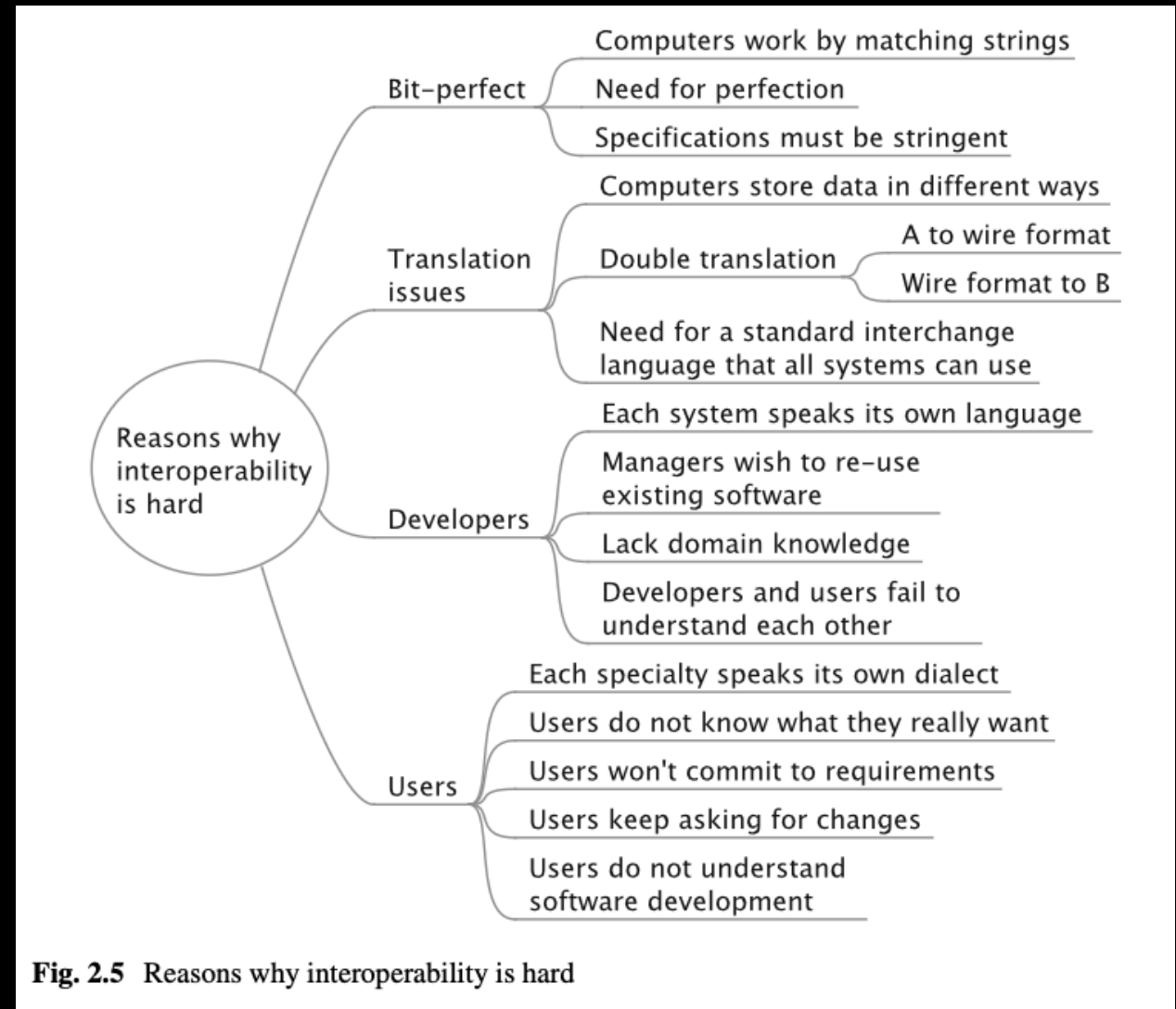


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